

Fast Multi-objective Evolutionary Algorithms to Approximate the Set of Optimal and Nearly Optimal Solutions

Carlos Segura , Angel E. Rodriguez-Fernandez , Carlos Hernández , Oliver Schütze 

Abstract—Recently, it has been argued that all nearly optimal solutions of a given multi-objective optimization problem (MOP) are of potential interest for the decision maker (DM). This set, $N_{Q,\epsilon}$, contains the entire Pareto set (and is thus interesting in the context of multimodal multi-objective optimization) as well as non- but nearly optimal solutions that can still serve as alternative or backup realizations. The DM is hence given an enlarged set of relevant possibilities.

In this work, we propose the first evolutionary algorithms aiming for finite-size approximations of the set of nearly optimal solutions of a given MOP. More precisely, we propose the two-phase framework TPEA- $N_{Q,\epsilon}$. In the first phase, an existing multi-objective evolutionary algorithm (MOEA) can be chosen to obtain a Pareto set approximation, while the second phase focuses on the detection of non- but nearly optimal solutions, with the novelty of using a clustering-based replacement strategy. Efficient algorithms to detect nearly optimal solutions of a discrete set for bi- and three-objective problems are proposed by combining specialized data structures and the two-pointer technique. A methodology to compare methods aiming to approximate $N_{Q,\epsilon}$ is defined. The strength of TPEA- $N_{Q,\epsilon}$ is demonstrated via empirical validation on several benchmark test problems.

Index Terms—Evolutionary computation, multi-objective optimization, nearly optimal solutions, fast $N_{Q,\epsilon}$ detection.

I. INTRODUCTION

MULTI-objective optimization problems (MOPs) naturally arise in many situations. For the treatment of such problems, multi-objective evolutionary algorithms (MOEAs) have been developed over the last three decades [1]–[3] and skillfully utilized to successfully solve a huge variety of applications (e.g. [4]–[6]). Until now, most MOEAs focus on Pareto front approximations. This is on the one hand of course justified since the Pareto front represents the set of optimal compromises of a given problem (in terms of the objective values). However, since decision-making processes are becoming more and more complex, researchers have identified other, enlarged sets of interest that may be helpful to the decision maker (DM). For instance, points on the Pareto front can have several pre-images. These pre-images represent

different alternative solutions to obtain the same quality (the latter again measured by the objective values). This calls for the detection of the Pareto set which is the main goal in multimodal multi-objective optimization (MMMO, [7]–[10]). Another flavor of MMO is to compute next to the Pareto set also all locally optimal solutions whose images are close to the Pareto front (e.g., [11], [12]). In [13], [14], it has been proposed to consider the set $P_{Q,\epsilon}$ of *approximate* solutions. This set contains, roughly speaking, all ϵ -efficient solutions (i.e., the solutions that are optimal when replacing the dominance relation via ϵ -dominance). $P_{Q,\epsilon}$, however, has the disadvantage that it contains the entire region around the hardly dominated boundary of a given MOP which is not usually desired by the DM, and which can lead to a computational burden. As a remedy, it has recently been proposed to consider the set $N_{Q,\epsilon}$ [15]. This set contains next to the entire Pareto set also non- but nearly optimal solutions that can serve as further interesting alternative realizations for the DM. $N_{Q,\epsilon}$ is hence interesting in the context of multimodal multi-objective optimization.

In [15], the set $N_{Q,\epsilon}$ and an unbounded archiver are proposed and analyzed. So far, however, no specialized solver exists that targets for $N_{Q,\epsilon}$ which is imperative to efficiently deal with nearly optimal solutions in applications. To fill this gap, we propose in this work TPEA- $N_{Q,\epsilon}$, a two-phase framework designed to compute suitable finite-size approximations of $N_{Q,\epsilon}$ of a given MOP. In the first phase, an existing MOEA can be chosen to obtain a Pareto set approximation, while the second phase focuses on the detection of non- but nearly optimal solutions. This stage includes a novel clustering-based replacement strategy aimed to increase the diversity of the resulting set. TPEA- $N_{Q,\epsilon}$ relies on detecting nearly optimal solutions in discrete sets. Specialized algorithms for bi- and three-objective problems to perform this step efficiently are defined. Finally, the strength of TPEA- $N_{Q,\epsilon}$ is demonstrated via empirical validation on several benchmark test problems, including two new problems and a practical one.

The main contributions of this work can be summarized as follows:

- Fast detection of $N_{D,\epsilon}$ for 2-D and 3-D finite sets with size N in asymptotically optimal time $O(N \log N)$. In contrast, the straightforward detection of $N_{D,\epsilon}$ has a complexity of $O(N^2)$.
- A two-stage algorithm (TPEA- $N_{Q,\epsilon}$) aimed at obtaining finite-size approximations of $N_{Q,\epsilon}$.
- A comparison with existing methods on benchmark problems.

The remainder of this work is organized as follows: in

C. Segura is with the Centre for Research in Mathematics (CIMAT), Computer Science Department, Callejón Jalisco s/n, Minetal de Valenciana, Guanajuato, Guanajuato 36240, Mexico

A. Rodríguez and O. Schütze are with the Computer Science Department, Cinvestav, Av. IPN 2508, C.P. 07360, Col. San Pedro Zacatenco, Mexico City, Mexico

C. Hernández is with the Computer Science Department, IIMAS-UNAM, Escolar 3000, C.U., Coyoacán, 04510 Ciudad de México, Mexico

O. Schuetze acknowledges support from Conahcyt (Secihti) project no. CBF2023-2024-1463. C. Hernández acknowledges support from PAPIIT UNAM project no. IA102025. A. Rodríguez acknowledges support from Secihti for his postdoctoral fellowship at Cinvestav.

Correspondence authors: Carlos Segura (carlos.segura@cimat.mx) and Oliver Schütze (schuetze@cs.cinvestav.mx)

Sec. II, we shortly recall the main definitions required to understand the sequel and discuss the related work. In Sec. III, we propose and discuss the algorithms for fast $N_{D,\epsilon}$ detection. In Sec. IV, we present the evolutionary framework TPEA- $N_{Q,\epsilon}$. In Sec. V, we present some numerical results that indicate the strength of the novel algorithms. A design problem concerning material selection is detailed and further explored in the supplementary material to highlight the practical significance of the $N_{Q,\epsilon}$ set. Finally, we draw our conclusions and give paths of possible future work in Sec. VI.

II. BACKGROUND AND RELATED WORK

We consider continuous multi-objective optimization problems (MOPs):

$$\min_{x \in Q} F(x). \quad (1)$$

F denotes the map that is defined by the k individual objectives f_i , $i = 1, \dots, k$, i.e., $F : Q \rightarrow \mathbb{R}^k$, $F(x) = (f_1(x), \dots, f_k(x))^T$. $Q \subset \mathbb{R}^n$ denotes the domain of F and the f_i 's, which can be expressed by equality and inequality constraints. n denotes the number of decision variables, and k the number of objectives. Given $v, w \in \mathbb{R}^k$, v is called less than w ($v <_p w$) if $v_i < w_i$ for all $i \in \{1, \dots, k\}$. $y \in Q$ is dominated by $x \in Q$ ($x \prec y$) wrt (1) if $F(x) \leq_p F(y)$ and $F(x) \neq F(y)$. $x \in Q$ is called a Pareto point or optimal if there does not exist a $y \in Q$ that dominates x . The set of all Pareto optimal points, P_Q , is called the Pareto set, and its image $F(P_Q)$ is called the Pareto front. Let $\epsilon = (\epsilon_1, \dots, \epsilon_k)^T \in \mathbb{R}_+^k$, then $x \in Q$ ϵ -dominates $y \in Q$ ($x \prec_\epsilon y$) wrt (1) if $F(x) - \epsilon \leq_p F(y)$ and $F(x) - \epsilon \neq F(y)$ [16], [17].

The concept of ϵ -dominance is used in [15] to define the set

$$N_{Q,\epsilon} := \{x \in Q \mid \exists p \in P_Q \text{ s.t. } x \prec_\epsilon p\} \quad (2)$$

of nearly optimal solutions within Q . More precisely, $N_{Q,\epsilon}$ contains all the solutions $x \in Q$ whose images $F(x)$ are “close enough” to the Pareto front (measured by ϵ). We stress that $P_Q \subset N_{Q,\epsilon}$ for all $\epsilon \in \mathbb{R}_+^k$. In other words, the entire Pareto set is always contained in the set of nearly optimal solutions. For graphical illustrations of $N_{Q,\epsilon}$ and a discussion of practical use cases, we refer the reader to [15]. Note that the additive ϵ -dominance concept is used to define $N_{Q,\epsilon}$. In principle, however, the multiplicative variant [18] can be used instead.

Evolutionary multi-objective optimization (EMO) has certainly been a story of great success for the last three decades [3]. During that time, many MOEAs have been proposed aiming for different sets of interest. By far most of these algorithms so far aim for suitable finite-size Pareto front approximations. These methods can mainly be divided into three classes: (i) MOEAs that are based on the concept of dominance (e.g., NSGA-II [19] and SPEA2 [20]), (ii) MOEAs that use decompositions (e.g., MOEA/D [21] and NSGA-III [22]), and (iii) MOEAs that utilize particular performance indicators (e.g., SMS-EMOA [23] and LIBEA [24] for the hypervolume indicator [25] and DDE [26] for the averaged Hausdorff

distance [27]). However, there have also been several attempts to compute further candidate solutions that may be of interest for the decision maker. ϵ -dominance has been used in [13], [14], [28] to capture the entire set of approximate solutions $P_{Q,\epsilon}$. This set, however, has the disadvantage that it includes all weakly optimal solutions, regardless of their distances to the Pareto front. It is unlikely that such points are relevant for the decision maker. To avoid this potential drawback, the set of nearly optimal solutions $N_{Q,\epsilon}$ has been proposed in [15]. Further, researchers have considered different sets of interest related to what is now known as multimodal multi-objective optimization (MMMO). The most common aim in MMO is to detect all pre-images of all points along the Pareto front (in other words, the entire Pareto set). Such attempts can be found e.g. in [7]–[10]. Landscape analysis aims in some cases to detect all locally optimal solutions – which includes all globally optimal ones – of a given problem, regardless of the distance of the images toward the Pareto front [29]–[32]. Recently, researchers have focused on the locally optimal solutions whose images are close to the Pareto front [11], [12]. Hereby, again, closeness is defined in objective space specified by a vector $\epsilon \in \mathbb{R}_+^k$. Finally, there are works on diversity optimization that not only focus on the Pareto front but also aim to obtain a diverse set of solutions in the decision space [33]–[35].

The fast detection of the promising solutions out of a given finite candidate set has ever been interesting in the EMO community as it has an impact on the speed of the algorithm. The most efficient algorithms for detecting the non-dominated solutions are $O(N \log^{k-2} N)$ for $k > 2$ (k the number of objectives and N the magnitude of the candidate set) and $O(N \log N)$ for $k = 2$. These algorithms usually rely on the divide and conquer approach [36] and data structures [37]. In the case of non-dominated sorting, additional steps are required. NSGA-II introduced a fast non-dominated sorting algorithm with a worst-case complexity of $O(kN^2)$ [19], which became a baseline for subsequent work. Jensen's divide-and-conquer approach reduced the worst-case complexity to $O(N \log^{k-1} N)$ by recursively sorting along the objectives [38], later extended by Fortin [39] to handle ties. More recent contributions, such as the divide-and-conquer algorithm [40] and the Merge Non-Dominated Sorting (MNDS) algorithm [41], achieved $O(N \log N)$ best-case complexity under typical assumptions. Refer to [42], [43] for an overview of MNDS and recent works in this field.

Clustering has been applied in various ways within MOEAs and more general in the context of multi-objective optimization. For instance, clustering can be used to support the selection mechanism in evolutionary algorithms [44]–[46]. In the context of archiving, it can be used to reduce large candidate sets (e.g., coming from external unbounded archivers) to a more manageable size for presentation to the decision maker [15], [47], [48]. Additionally, clustering has been employed as a subset selection method [49], [50], particularly k -means [51], since its objective function coincides with the IGD indicator [50], [52].

Algorithm 1 Fast $N_{D,\epsilon}$ detection for bi-objective problems

```

1: Input: discrete set  $D \subset \mathbb{R}^n$  with no identical objective vectors,
   threshold  $\epsilon \in \mathbb{R}_+^k$ 
2: Output:  $N_{D,\epsilon}$ 
3:  $N_{D,\epsilon} = \emptyset$ 
4:  $\mathbf{P}_D := \text{Non-Dominated}(D)$ : Use any  $O(|D| \log |D|)$  algorithm.
5: Sort  $\mathbf{P}_D$  lexicographically: sort by first objective.
6: for all  $d \in D$  (in any order) do
7:   Apply binary search to find  $p \in \mathbf{P}_D$  minimizing  $f_1(p)$  s.t.
      $f_1(p) \geq f_1(d) - \epsilon_1$ 
8:   if ( $p$  exists) and ( $d \prec_\epsilon p$ ) then
9:      $N_{D,\epsilon} := N_{D,\epsilon} \cup \{d\}$ 
10:  end if
11: end for
12: return  $N_{D,\epsilon}$ 

```

III. FAST DETECTION OF NEARLY OPTIMAL SOLUTIONS

Here we present two algorithms for the fast detection of the nearly optimal solutions out of a finite set D of candidate solutions. That is, the domain is given by D in this section, and we hence use the notation $N_{D,\epsilon}$. The first algorithm is designed for bi-objective problems and the second one for three-objective problems. The complexity in both cases is $O(N \log N)$, where $N = |D|$, i.e., it shares the complexity with the best-known algorithms for calculating the set of non-dominated solutions [36]. However, the algorithm and data-structures required are more complex in this case.

a) Fast $N_{D,\epsilon}$ detection for bi-objective problems: In the two-objective case, the ideas behind the efficient solvers are somewhat similar to those used for calculating the non-dominated solutions [53]. First, the non-dominated set P_D out of D is calculated. This can be done with any of the $O(|D| \log |D|)$ algorithms that have already been developed. Then, all the solutions in P_D are sorted lexicographically, e.g., wrt the first objective. The solutions $d \in D$ are traversed in any order. As described above, a solution $d \in D$ is contained in $N_{D,\epsilon}$ if it ϵ -dominates any of the solutions $p \in P_D$. Since P_D is sorted lexicographically, binary search can be used to efficiently detect this situation: given $d \in D$ with objective values $F(d) = (f_1(d), f_2(d))^T$, binary search is used to identify the solution p in P_D whose objective value $f_1(p)$ is minimal subject to the constraint that this value is higher or equal than $f_1(d) - \epsilon_1$. The solution d belongs to $N_{D,\epsilon}$ if and only if p exists and if d ϵ -dominates p . Else, d does not belong to $N_{D,\epsilon}$. See Alg. 1 for the pseudocode of the algorithm.

b) Fast $N_{D,\epsilon}$ detection for three-objective problems: A more complex algorithm is proposed for three-objective problems. The goal is to identify the solutions $d \in D$ that ϵ -dominate at least one solution in P_D .

The difficulty in this case is that even if P_D is detected and sorted lexicographically, it is not easy to efficiently detect if there exist a solution $p \in P_D$ that is ϵ -dominated by a given solution $d \in D$. Differently to the bi-objective case, the solution $p \in P_D$ that minimizes its first objective, subjected to the constraint that it is higher or equal than $f_1(d) - \epsilon_1$, might not be ϵ -dominated by d but an alternative solution might be ϵ -dominated. Thus, a different strategy is used. Particularly, we propose to use the methodology known as the two-pointer

Algorithm 2 Fast $N_{D,\epsilon}$ detection for three-objective problems

```

1: Input: discrete set  $D \subset \mathbb{R}^n$  with no identical objective vectors,
   threshold  $\epsilon \in \mathbb{R}_+^k$ 
2: Output:  $N_{D,\epsilon}$ 
3:  $N_{D,\epsilon} := \emptyset$ 
4:  $\mathbf{P}_D := \text{Non-Dominated}(D)$ : Use any  $O(|D| \log |D|)$  algorithm.
5: Apply coordinate compression: transform the second objective of
   all  $d \in D$  to natural numbers while preserving the original order.
   The sub-index  $2^*$  is used to denote the objective transformed.
6: Sort  $\mathbf{P}_D$  and  $\mathbf{P}_D$  lexicographically in ascending order: sort by
   first objective, in case of tie by second objective and in case of
   tie by third objective.
7:  $\mathbf{st} := \text{Segment\_Tree\_Init}(|D|, -\infty)$ : initialize the segment
   tree with  $|D|$  positions equal to  $-\infty$ .
8:  $\mathbf{NSI} := |\mathbf{P}_D|$ .  $\mathbf{NSI}$  stands for the Next Solution to Insert in the
   segment tree (two-pointer technique).
9: for  $d \in D$  in decreasing lexicographical order do
10:  while ( $(\mathbf{NSI} \geq 1)$  and  $(f_1(P_D[\mathbf{NSI}]) \geq f_1(d) - \epsilon_1)$ ) do
11:    Update segment tree  $\mathbf{st}$ : update the position  $P_D[\mathbf{NSI}]_{2^*}$ 
      of  $\mathbf{st}$  to the maximum between the current value and
       $P_D[\mathbf{NSI}]_3$ 
12:     $\mathbf{NSI} := \mathbf{NSI} - 1$ 
13:  end while
14:  if  $d \in P_D$  then
15:     $N_{D,\epsilon} := N_{D,\epsilon} \cup \{d\}$ 
16:  else
17:     $c := \text{binary\_search\_obj2}(f_2(d) - \epsilon_2)$ 
18:    if  $(f_3(d) - \epsilon_3 \leq \text{st.max\_range}(c, |D|))$  then
19:       $N_{D,\epsilon} := N_{D,\epsilon} \cup \{d\}$ 
20:    end if
21:  end if
22: end for
23: Erase individuals on the boundary: erase all solutions  $d \in N_{D,\epsilon}$ 
   with  $F(d) - \epsilon \in P_D$ .
24: return  $N_{D,\epsilon}$ 

```

technique in combination with a segment tree [54]. The general idea is to traverse the solutions $d \in D$ in a specific order while maintaining a data structure (segment tree) that contains only solutions in P_D that might be ϵ -dominated by d . This data structure allows one to efficiently detect if an ϵ -dominated solution exists. The computation is organized in a way that each solution in P_D is inserted into the data structure once, which ensures efficiency. See Alg. 2 for the pseudocode of the algorithm. As in the bi-objective case, the first step is to calculate P_D . This can be done with any of the existing $O(|D| \log |D|)$ algorithms. Additionally, the coordinate compression technique is used to transform the second objective to natural numbers [55]. Note that coordinate compression is the process of mapping each value in a list to its index if that list was sorted. Thus, the values of this objective are temporarily changed to natural numbers in $[0, N-1]$ preserving the original order. This step is required to use the second objective to index the segment tree. In Alg. 2 the sub-index 2 is used to refer to the original second objective, whereas 2^* is used to refer to the objective after the coordinate compression. The solutions $d \in D$ are sorted in lexicographical order and traversed in decreasing order. If d belongs to P_D it is inserted into $N_{D,\epsilon}$. Otherwise, it must be checked if there exists a solution $p \in P_D$ such that $d \prec_\epsilon p$. Note that only the solutions that fulfill $f_1(p) \geq f_1(d) - \epsilon_1$ can be ϵ -dominated by d . All those solutions are inserted into a segment tree. Specifically, the

second objective — previously converted to a natural number with coordinate compression — is used as the index, while the third objective is the content of the leaf nodes of the segment tree. The segment tree is used to detect if any of the inserted solutions is ϵ -dominated by d . This can be done in $O(\log |D|)$ with a range maximum query. Particularly, we need to know if there exists a solution p in the segment tree that fulfills $f_2(p) \geq f_2(d) - \epsilon_2 \wedge f_3(p) \geq f_3(d) - \epsilon_3$. The condition $f_2(p) \geq f_2(d) - \epsilon_2$ is used to detect the appropriate index for the range maximum query. Particularly, a binary search is done on the values of the second objective (previously sorted) to detect the minimum index (c) with a value larger than or equal to $f_2(d) - \epsilon_2$. Then, the range associated to the query is $[c, |D|]$. If d belongs to $N_{D,\epsilon}$ the result of the range maximum query must be larger than or equal to $f_3(d) - \epsilon_3$. Finally, note that, since the solutions $d \in D$ are traversed in lexicographical decreasing order, this means that the solutions p that need to be considered is a superset of the ones considered for the previous solution checked in D . For this reason, the two-pointer technique can be used to insert each solution at most one time in the segment tree, meaning the total complexity is equal to $O(|D| \log |D|)$. See lines 10-13 for the details of the usage of the two-pointer technique. Note that in order to compare the solutions the relation greater than or equal to was considered in all cases. This means that a solution that is equal to a non-dominated solution plus ϵ is accepted. However, these solutions are not part of $N_{D,\epsilon}$. Thus, the last step of Alg. 2 is to erase those solutions (line 23).

As a summary, some of the most important features and components of the approach are the following:

- 1) The non-dominated set P_D is identified in $O(|D| \log |D|)$ with standard algorithms.
- 2) Solutions are sorted lexicographically by (f_1, f_2, f_3) to allow the use of two-pointer sweep.
- 3) Coordinate compression is applied to the f_2 values to operate on integer indices.
- 4) A structure indexed by f_2 (post-compression) stores associated f_3 values and allows efficient range maximum queries.
- 5) Binary search is used to identify the querying range.
- 6) Two-pointer sweep is used to insert each non-dominated vector only once and to identify the candidates of interest.

We stress that the previous algorithm descriptions assume no two solutions in D have identical objective vectors. In order to avoid issues when solutions with identical vectors are present, they must be temporarily removed prior to the application of the previous algorithms. If a solution belongs to $N_{D,\epsilon}$ all solutions with identical objective vectors also belong to the set, so a post-process to insert them is used.

In some cases complexity analysis is a bit misleading in the sense that the hidden constants are not taken into account. In order to show that the designed algorithms are practical, it was applied to sets D with up to 2,000,000 vectors both for the two-dimensional and three-dimensional case. Additionally, a quadratic algorithm to detect $N_{D,\epsilon}$ was also implemented, which operates as follows. First, the non-dominated set is

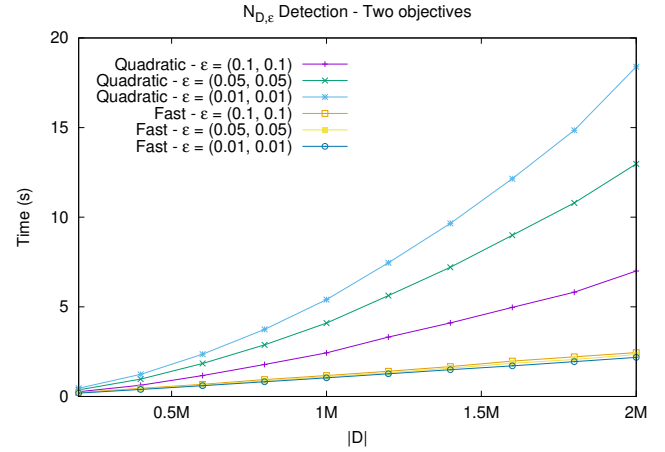


Fig. 1. Time required to calculate $N_{D,\epsilon}$ using fast detection and quadratic algorithm for the two-objective cases.

calculated in $O(|D| \log |D|)$. Then, for each individual not belonging to the non-dominated set, it checks if it ϵ -dominates any of the non-dominated individuals to properly identify $N_{D,\epsilon}$ members, by just following the definition. Note that elements in the non-dominated set are considered in a random order, which is important to improve the performance. The worst case appears when there are many solutions that belong to the non-dominated front and many solutions that do not belong to $N_{D,\epsilon}$. In particular, if $|D|/2$ solutions are non-dominated and $|D|/2$ solutions are not part of $N_{D,\epsilon}$, all these solutions are pair-wise compared. This kind of case was not considered in our validation since it is probably a pathological case not appearing in practical situations.

In our validation, vectors $s \in D$ were created following the next random process: $s_1 \sim U(0, 1)$, $s_2 \sim -s_1 + U(0, 0.3)$, $s_3 \sim -s_1 + U(0, 0.3)$. Three different values of ϵ were tested to generate cases where the amount of vectors belonging to $N_{D,\epsilon}$ varies. In the cases of bi-objective vectors they were: $(0.01, 0.01)^\top$, $(0.05, 0.05)^\top$ and $(0.1, 0.1)^\top$, and analogously for three-objective vectors. With the different ϵ values and normalized objectives, the percentage of vectors that belong to $N_{D,\epsilon}$ is roughly in the range [7.5, 15]% for 0.01, [35, 65]% for 0.05 and [75, 95]% for 0.1. In order to perform the time-analyses, tests were repeated 10 times with different seeds. Experiments were carried out in a cluster whose nodes have two Intel Xeon E5 v4 family processors and 32 GB of RAM. Fig. 1 shows the mean time used to find $N_{D,\epsilon}$ for the different ϵ values and tested algorithms in the two-objective case. The same information is shown in Fig. 2 for the three-objective case.

As expected, in the fast $N_{D,\epsilon}$ detection algorithm the time increases slightly faster than linear both in the two-objective and three-objective cases and, as ϵ increases, the time grows slightly since $N_{D,\epsilon}$ is larger. However, for most practical cases, these times will probably be negligible since working with sets of 2,000,000 members and the largest ϵ takes less than 10 seconds for the three-objective cases and less than 2.5 seconds for the two-objective cases. Note that even with cases that were not created to harm the performance

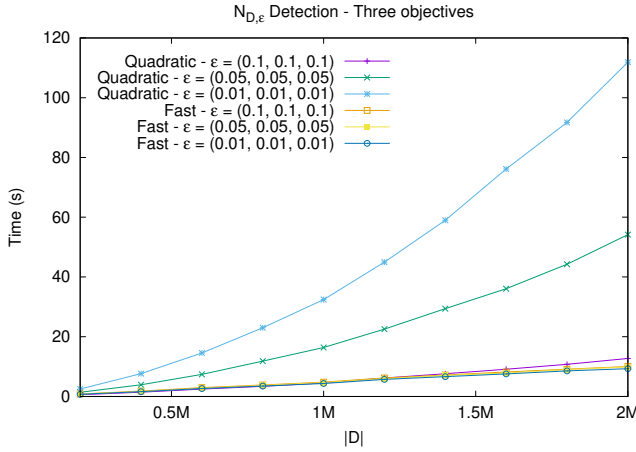


Fig. 2. Time required to calculate $N_{D,\epsilon}$ using fast detection and quadratic algorithm for the three-objective cases.

of the quadratic algorithm, the experiments clearly show the important advantages of the fast $N_{D,\epsilon}$ detection algorithms, with an increase in time provoked by the quadratic algorithm of more than 750% in the case of $\epsilon = (0.01, 0.01)^\top$ and more than 1000% in the case of $\epsilon = (0.01, 0.01, 0.01)^\top$. The experiments on the test problems used in Section V show a similar impact on the performance. The reason for the low performance of the quadratic algorithm is that the amount of individuals not belonging to $N_{D,\epsilon}$ is large, meaning that they have to be compared against all the non-dominated solutions to confirm that they do not belong to $N_{D,\epsilon}$.

Also note that for very high values of ϵ , the advantages of fast $N_{D,\epsilon}$ detection diminishes for the three-objective case. The reason is that in this case most of the individuals belong to $N_{D,\epsilon}$ and for each individual that is dominated, it is easy to find a non-dominated vector that it ϵ -dominates. This is probably also a pathological case in practical situations, since using a large ϵ value with respect to the range of the objectives can generate huge $N_{D,\epsilon}$ sets with not so high-quality solutions (owed to the large values of ϵ). However, even in these cases, using fast $N_{D,\epsilon}$ detection does not harm the performance. Thus, the proposals put forth in the next section make use of the fast $N_{D,\epsilon}$ detection whenever this set needs to be calculated.

IV. AN EVOLUTIONARY FRAMEWORK FOR THE APPROXIMATION OF $N_{Q,\epsilon}$

Here we propose TPEA- $N_{Q,\epsilon}$ (Two-Phase Evolutionary Algorithm for $N_{Q,\epsilon}$), the first evolutionary framework that aims to compute finite-size approximations of the set of nearly optimal solutions of a given MOP. Alg. 3 shows the pseudocode of this algorithm¹.

In order to approximate $N_{Q,\epsilon}$, a proper approximation of the Pareto set P_Q is required. We have hence opted to divide the algorithm into two phases. The main aim of the first phase (lines 3-4) is to attain a proper estimate of P_Q , whereas the second phase (lines 5-8) focuses on extending the set identified

in the first phase to properly approximate $N_{Q,\epsilon}$. In principle, any existing MOEA might be used in the first phase (line 4). Note, however, that $N_{Q,\epsilon}$ is a superset of P_Q and that diversity in decision variable space is desired in order to present the DM interesting back-up or alternative solutions. If points on the Pareto fronts have multiple pre-images, P_Q will likely consist of different connected components. Elitist MOEAs as NSGA-II and MOEA/D may not sample or even abandon such regions since their primary goal is to get an approximation of the Pareto front. Hence, such algorithms might not be the best choice for the first phase.

In order to take the aforementioned issues into account, two design decisions were taken. First, the starting set of the second phase is not the final population of the first phase. Instead, all the solutions evaluated during the first phase are considered to get an initial approximation of $N_{Q,\epsilon}$ with normalized objectives (line 6) using the fast $N_{Q,\epsilon}$ algorithm presented in Sec. III and then a bounded selector based on clustering principles is used to get a bounded but diverse approximation with N_2 elements (line 7) that is used as the initial population of the second phase. The clustering-based bounded selector (Alg. 4) is explained below. Note that using all the solutions evaluated during the first phase to get an approximation of $N_{Q,\epsilon}$ (line 6) might appear to be very costly. However, for discrete sets with N elements, the cost is $O(N \log N)$ both for two- and three-objective problems, so this part was not computationally heavy (although it requires to store all the solutions in memory). Second, the experimental validation does not only consider elitist MOEAs for the first phase. There are several MOEAs that have already explored the advantages of explicitly managing the diversity of the variable space to alter the features of the output in terms of increasing diversity or quality. These ideas have been explored mainly in Multimodal Multi-Objective Evolutionary Algorithms. Note that similarly to the proposed MOEA, these algorithms aim to increase the diversity in the variable space. However, they do not explicitly manage an acceptable worsening through the formalization of the set $N_{Q,\epsilon}$. In any case, the ideas explored in such methods seem promising for approximating $N_{Q,\epsilon}$. VSD-MOEA attained impressive results in complex benchmarks with two and three objectives. The main feature of VSD-MOEA [56] is that it introduces a novel replacement strategy that explicitly induces a slow decrease of the diversity with the aim of avoiding premature convergence. We have hence chosen to utilize this MOEA in the first phase.

In the second phase, the bounded approximation of $N_{Q,\epsilon}$ mentioned before is used as the initial population of an EA (line 8). This EA follows the general flowchart of traditional EAs. In particular, it is an iterative procedure that applies parent selection, reproduction and replacement. No novelties were included in the parent selection and reproduction phase. Parents are selected and paired randomly, and SBX crossover and polynomial mutation were applied to create as many offspring as individuals in the population. Any other mechanisms might be readily applied in these phases. The main novelty appears in the replacement phase. First, the members of the population and offspring are joined. Then, the fast $N_{Q,\epsilon}$ detection algorithm is applied considering normalized objectives. If

¹Source code can be downloaded at <https://zenodo.org/records/20848264>

Algorithm 3 TPEA- $N_{Q,\epsilon}$

```

1: Input:  $N_1, N_2$  (population size in first and second phase),
    $\epsilon$  (threshold used for the definition of  $N_{Q,\epsilon}$ ),  $tdo$  (threshold
   distance used for the bounded selector),  $VP$  (variables to pro-
   mote in bounded selector),  $Stop_1, Stop_2$  (budget of function
   evaluations).
2: Output: FinalApprox (discrete approximation of  $N_{Q,\epsilon}$ )
3: //First phase
4:  $AllInd = MOEA(N_1, Stop_1)$ : Apply existing MOEA with
   population size  $N_1$  for  $Stop_1$  evaluations. It returns a set with
   all the solutions evaluated.
5: // Second phase
6:  $NQ' := Fast - N_{Q,\epsilon}(AllInd, \epsilon)$  // Algorithm 1, 2 with
   normalized objectives
7:  $BoundedApprox = BoundSelect(NQ', N_2, VP, tdo)$  //
   Algorithm 4.
8:  $FinalApprox := Clustered-N_{Q,\epsilon}$ -EA( $BoundedApprox, N_2,$ 
    $Stop_2, \epsilon, tdo, VP$ ): it incorporates Algorithm 4 as survivor
   selection strategy.
9: return  $FinalApprox$ 

```

Algorithm 4 Bounded Selector for $N_{Q,\epsilon}$

```

1: Input:  $NQ'$  (candidate set),  $lim$  (maximum size of approxima-
   tion),  $VP$  (set of variables to promote),  $tdo$  (threshold distance
   in objective space).
2: Output: SA (approximation of  $N_{Q,\epsilon}$ )
3: if  $|NQ'| \leq lim$  then
4:   return  $NQ'$ 
5: end if
6:  $P_{NQ'} := \text{Non-Dominated}(NQ')$ 
7:  $A = \{b_j | b_j \text{ is the best solution in } P_{NQ'} \text{ for objective } j \in \{1 \dots k\}\}$ 

8: //Selections focused on objectives
9: while  $(|A| \leq lim)$  do
10:    $far := \arg \max_{x \in P_{NQ'}} ObjClosestNormD(x, A)$ 
11:   if  $ObjClosestNormD(far, A) \geq tdo$  then
12:      $A := A \cup \{far\}$ 
13:   else
14:     break
15:   end if
16: end while
17: //Selections focused on objectives and  $VP$ 
18:  $R := NQ' \setminus A$ 
19: for all  $a_i \in A$  do
20:   Create set  $SA_i := \{a_i\}$ 
21: end for
22: for  $r \in R$  do
23:   Make  $s(r)$  the set  $SA_i$  with the closest member to  $r$  consid-
   ering normalized objective distance
24: end for
25: while  $\sum(|SA_i|) \leq lim$  do
26:    $far := \arg \max_{r \in R} VarClosestNormD(r, s(r), VP)$ 
27:    $s(r) := s(r) \cup \{far\}$ 
28:    $R := R \setminus \{far\}$ 
29: end while
30:  $SA := \bigcup_{i=1}^{|A|} SA_i$ 
31: return  $SA$ 

```

the resulting set is shorter than N_2 — which is both the desired size of the output and population size of the EA — all these elements survive to the next generation. Otherwise, the same clustering-based bounded selector approach (Alg. 4) that was used to select the initial population of the second phase is applied to limit the size of the set to N_2 .

Regarding the bounded selector based on clustering we

remark that it is used in two different parts. First, it is used to create the initial population of the second phase. Second, it is used in the replacement strategy of the second phase. This algorithm receives as input a set of solutions (NQ') whose magnitude may be larger than N_2 . In both cases, these solutions are the output of the fast $N_{Q,\epsilon}$ detection algorithm. The aim is to generate a bounded set limited to N_2 members that represent a good approximation of P_Q and $N_{Q,\epsilon}$. In particular, we are interested in maintaining for each region of P_Q distant solutions in the search space. Thus, it makes sense to maintain two similar solutions in the variable space if they are distant in the objective space, as well as two similar solutions in the objective space if they are distant in the variable space. However, maintaining pairs of solutions that are similar both in the objective and the variable space is not that helpful for the decision makers. The guiding principle of the approach is to first cluster solutions by their positions in the objective space, and then maximize variable space diversity at each region.

First, Alg. 4 focuses on obtaining a proper discretization of the non-dominated solution of NQ' attending to the objective space (lines 6 - 16). This set is called A . Initially, after calculating the non-dominated individuals ($P_{NQ'}$ - line 6), the best solutions in $P_{NQ'}$ for each of the objectives are selected to fill A (line 7). The idea behind this selection is to promote the further extension of $P_{NQ'}$. Then, for each solution $y \in P_{NQ'}$ we maintain the distance to its closest solution in A . Note that objectives are normalized to avoid biasing the search towards some particular objectives if they have different ranges. Iteratively, we select the solution $x \in P_{NQ'}$ with the largest distance (lines 9-16). This process finishes when the maximum distance is lower than a given threshold value (tdo) which should be set by the DM. The guiding principle is that solutions with distances lower than tdo should be considered similar by the DM, so further selection by this criterion should not be carried out. Note that since the resulting set must be lower than N_2 , if the set A reaches N_2 solutions, the process also stops. However, in our experimental validation tdo and N_2 were set so that this process stops before filling A with N_2 solutions. In particular, in most cases, it terminates with around $0.1 \times N_2$ solutions, meaning that several of the solutions are integrated by considering the next steps, which also consider the diversity in the variable space.

The second part (lines 17-29) focuses on maximizing the variable-space diversity for each of the regions identified. Note that since $N_{Q,\epsilon}$ can be quite large, DMs might select a subset of variables to promote the diversity. This set of variables to promote is called VP in the pseudocode. In order to promote the diversity in those solutions, we initially create for each $a_i \in A$ a set whose only member is a_i (lines 19-21). Thus, we manage $|A|$ different sets. Each of the solutions that are not currently in any of the sets ($R := NQ' \setminus A$) are associated to one set. In particular, each solution $r \in R$ is associated to the set with the closest $a \in A$ and the corresponding set is called $s(r)$ (lines 22-24). Normalized distances in the objective space are used. Thus, for each solution we identify its closest region in A . Then, at each step, we select a solution in R to insert it to its associated set. This is done by considering the one

with the maximum search-space diversity contribution (lines 25-29). This contribution is calculated by considering for each $r \in R$ the distance to the closest solution in $s(r)$. This distance is calculated with a normalized variable-space distance that takes into account only the variables in VP . The final output is the union of all the generated sets and the process stops when the cardinality is equal to lim .

V. EXPERIMENTAL VALIDATION

The aim of this section is to show the effectivity of the method put forth in this paper and to better understand the effects of its components. In order to perform the comparisons, four variants are analyzed using different components in $TPEA-N_{Q,\epsilon}$. For the first stage, two different MOEAs are used: NSGA-II and VSD-MOEA. NSGA-II is the most well-known elitist MOEA, whereas VSD-MOEA introduces explicit management of diversity, which is one of the main components used in multimodal optimization. For each of them, two variants are considered by activating or deactivating the second stage. For the cases in which the second stage is deactivated, using the set of non-dominated solutions obtained by the algorithm as the output would be unfair. Thus, in every method all the solutions evaluated during the optimization process are stored. Then, at the end of the optimization, N_2 members were selected by first applying fast $N_{Q,\epsilon}$ detection with normalized objectives followed by Algorithm 4. Note that in the case of using two phases in $TPEA-N_{Q,\epsilon}$, this set is quite similar to $FinalApprox$ in Alg. 3. However, even in this case we use the same approach to ensure fair comparisons. Also note that, although we use long-term experiments, the efficient components designed in this paper allow us to operate with the whole set of solutions both in the two-dimensional and three-dimensional cases with a reduced computational burden. In addition, HREA [11] is also used to validate the advances. To ensure fair comparisons, the final solution of HREA is generated in the same way, i.e. using fast $N_{Q,\epsilon}$ detection and Algorithm 4 to select N_2 solutions.

Four experiments were performed. The first one uses two newly defined simple problems, as well as some additional problems that arose in MIMO. The experimentation sheds light on the main benefits contributed by the second phase of $TPEA-N_{Q,\epsilon}$ and by the explicit management of diversity. This first analysis is done in the long-term, while the second experiment analyzes performance under more restricted computational budgets. The third experiment considers the WFG problems to analyze the contributions of $TPEA-N_{Q,\epsilon}$ for more complex problems. Finally, a practical problem is used to show the effects of ϵ and tdo .

A. Methodology

Given the novelty of the formulation presented in this paper, it is important to define a methodology to make fair comparisons among different $N_{Q,\epsilon}$ approximations. $TPEA-N_{Q,\epsilon}$ receives two parameters specific to this set: the value of ϵ and the set of variables where diversity should be promoted (VP). Then, the aim is to attain a bounded approximation of $N_{Q,\epsilon}$ that maximizes the diversity in the selected variables.

Note that for many problems, individuals with images that are on the boundary of $F(N_{Q,\epsilon})$, i.e. in the objective space, are also on the boundary of $N_{Q,\epsilon}$, i.e. in the variable space. It is hence expected that individuals on the boundary are generated by $TPEA-N_{Q,\epsilon}$ and other solvers aimed to attain a bounded approximation of $N_{Q,\epsilon}$ with large diversity. If a method does not exactly find the Pareto front but has a minor convergence gap — which is the expected behavior — the solutions that are identified as part of the boundary in such an execution might be very close to the $N_{Q,\epsilon}$ set but out of it. Totally disregarding those solutions would result in unfair comparisons.

Given a set of solutions S generated by a method, the first step in our methodology is to filter out the solutions that do not belong to $N_{Q,\epsilon}$. However, the value ϵ used to filter out the solutions is slightly increased with respect to the value used as input to $TPEA-N_{Q,\epsilon}$ to avoid the effect previously described. In particular, in this paper, the values of ϵ are multiplied by $\Theta = 1.2$.

Since the Pareto front is generally unknown, another important aspect is how to determine if a solution belongs to the set of interest or not. In this paper, we use problems where the Pareto front is known, so this knowledge might be used. However, since some complex problems that are not usually solved to optimality by MOEAs are considered, using the non-dominated front would provoke incomparable results, since just a few or even no points of $N_{Q,\epsilon}$ would be found. Thus, we approximate the non-dominated front with the results obtained during the experimentation. All the solutions generated in any of the experiments are joined, and the set of non-dominated solutions is identified. This is called the whole set (WS). A solution x obtained by any method is filtered out if it does not ϵ -dominate (with the value of ϵ multiplied by Θ and normalized objectives) any of the solutions of WS . Given a solution set identified by a method, the set generated after filtering out solutions is called the approximate set (AS).

In addition, to calculate the indicator $MSTI$, which is explained below, the set WS is reduced to contain only R members using the following procedure. The first member is selected at random. Then, for $R - 1$ iterations, the distance (in normalized objective space) of each non-selected solution to the closest selected solution (the ones already in the set) is calculated and the one with the largest distance is moved to the selected set. The reduced set is called the reference set (RS).

Finally, we describe some strategies for measuring the quality of AS . It is important to use indicators that assign a real value to any candidate set AS with the aim of facilitating comparisons among methods. Given a set of non-dominated vectors, in [15] it is proposed to detect the set of close solutions to each vector. Further, it is proposed to use the mean length of the minimum spanning trees (MST) covering those solutions to summarize the quality of the approximation. The normalized distances between solutions for the MST are calculated taking into account the variables in VP . In order to take into account solutions in different parts of the front, the set RS is considered. Each solution in AS is matched to the closest solution in RS . The normalized Euclidean distance in objective space is considered to do the matching. Then, for each $s \in RS$, the set of solutions matched to it is used to

generate the MST, and their lengths are averaged to summarize the accuracy of *AS*. This indicator is called the Minimum Spanning Tree Indicator (*MSTI*). In our validation, the size of *RS* is set to 30 for problems with two objectives and to 750 for problems with three objectives. We emphasize that while *MSTI* measures diversity, the solutions in the approximate set (*AS*) are filtered by quality; thus, the indicator incorporates both dimensions.

MSTI allows to summarize the quality of an approximation with a real number, meaning we can use any of the statistical tests typically applied to compare quality in approximate optimization. The guideline described in [57] is used to statistically compare the results of different methods. First, the Kruskal-Wallis test was used as an omnibus test to determine whether significant differences existed between algorithms. If significant differences exist, the Mann-Whitney test with Hommel's correction was performed for pairwise comparisons, ensuring control over the Type I error rate. A significance level of 0.05 was used.

In addition to designing indicators, visualizations are also important. Two natural visualizations are the following. First, since the experimental validation in this paper promotes the diversity in at most two variables, the set *AS* that obtains the median *MSTI* value can be plotted by considering only the variables of interest. While the previous plot offers a visualization of the set of regions globally covered by a typical execution of the method, it can be misleading because most of the diversity might be associated to just a few regions of the Pareto front. Thus, if the decision maker selects a different region, it might not have enough alternatives. In order to illustrate the variability obtained for a region of the non-dominated front, a more informative plot is the spanning tree that obtains the median length of the execution with median *MSTI*. Note that if several results are plotted simultaneously, drawing the edges might saturate the figure. Thus, an alternative is to plot only the set of vertices. We focus our visualizations on this second kind of plot, which we call *MSTML* plots (Minimum Spanning Tree with Median-Length).

B. Contribution of the main components of *TPEA-N_{Q,ε}*

As mentioned previously, this experimental validation considers four configurations to perform an ablation study where both the second phase and the explicit management of diversity are activated or deactivated. In order to denote the four configurations, we use the following tags: *NSGA-II*, *VSD-MOEA*, *NSGA-II-2S*, *VSD-MOEA-2S*, where the suffix *2S* means that the second stage is activated. Note that all of them are realizations of *TPEA-N_{Q,ε}*. Additionally, *HREA* is used to validate the advances provided by *TPEA-N_{Q,ε}*.

This section considers two newly defined problems and five problems that have been proposed in order to validate advances in multimodal multi-objective optimization. The two new problems were specifically designed to better show the logic behind including explicit management of diversity and the second stage in *TPEA-N_{Q,ε}*. The problem *DUNQ* stands for distant unconnected *N_{Q,ε}*. *DUNQ* consists of two variables (x_1 and x_2) and two objective functions (*DUNQ₁* and *DUNQ₂*) as given in (3):

$$\begin{aligned} DUNQ_1(x) &= (r(x_1)\%64) + g_1(x_1 - 4r(x_1) - 2, x_2) \\ DUNQ_2(x) &= (r(x_1)\%64) + g_2(x_1 - 4r(x_1) - 2, x_2) \\ \text{where} \\ r(v) &= \lfloor v/4 \rfloor \\ g_1(z) &= z_1^4 + z_2^4 - z_1^2 + z_2^2 - 10z_1z_2 + 0.25z_1 + 20 \\ g_2(z) &= z_1^2 + (z_2 - 1)^2 \end{aligned} \quad (3)$$

The bound constraints are $x_1 \in [0, 1536]$ and $x_2 \in [-2, 2]$. For small values of ϵ , $N_{Q,\epsilon}$ falls into six disconnected regions. We note that the problem is generated by replicating variants of the landscape generated by g_1 and g_2 , which always receive both parameters in the range $[-2, 2]$ independently of the value of x_1 and x_2 . This is done through a periodic transformation. In addition, the *DUNQ* functions introduce a penalty based on a modular operation, so six distant high-quality regions appear in the variable space. Since the regions appearing in $N_{Q,\epsilon}$ are distant and disconnected, promoting a slow convergence through the explicit management of diversity in the first phase seems promising to locate the different regions.

The second benchmark is *LCNQ* which stands for Large Connected $N_{Q,\epsilon}$. *LCNQ* consists of 24 variables and two objective functions (*LCNQ₁* and *LCNQ₂*) as given in (4):

$$\begin{aligned} LCNQ_1(x) &= \sum_{i=3}^{24} 10^{-4} x_i + g_1(x_1, x_2) \\ LCNQ_2(x) &= \sum_{i=3}^{24} 10^{-4} x_i + g_2(x_1, x_2) \\ \text{where} \end{aligned} \quad (4)$$

$$\begin{aligned} g_1(z) &= \begin{cases} 1 - z_2 & \text{if } z_1 + z_2 < 1 \\ z_1 & \text{else} \end{cases} \\ g_2(z) &= \begin{cases} 1 - z_1 & \text{if } z_1 + z_2 < 1 \\ z_2 & \text{else} \end{cases} \end{aligned}$$

The bound constraints are $x_i \in [0, 1]$ for all the variables. We stress that the variables $x_3, \dots, x_{24}$ have a reduced effect on the quality because these values are multiplied by 10^{-4} prior to summing them up to the objective functions. Thus, the set $N_{Q,\epsilon}$ is huge. In fact, with the threshold used in the experimentation, $\epsilon = (0.01, 0.01)^\top$, there are solutions in $N_{Q,\epsilon}$ even with the maximum values in such variables.

In order to perform the validation, standard components and parameter values are considered in all the evolutionary algorithms. In particular, the parameterization given in Table I is used. For *NSGA-II-2S* and *VSD-MOEA-2S*, *Stop₁* was set to the 60% of the total number of evaluations, meaning that the second stage uses 40% of the computational resources. For *VSD-MOEA*, its internal parameter *ITV* was set to 0.4 as recommended by the authors. These settings are maintained in every experiment unless otherwise stated. Note that when the second phase is deactivated, *Stop₁* is set to the 100% of the total number of evaluations. In the case of *DUNQ*, the stopping criterion was set to 60,000 evaluations and the output size (N_2) to 600 members. That is, ideally, each of the regions of the non-dominated front (30) should be covered by 20 solutions. The variables to promote (*VP*) are x_1 and x_2 . Since

TABLE I
PARAMETERIZATION COMMON TO EVERY CONFIGURATION

Parameter	Value
Population size (N_1)	100
Crossover	SBX
Mutation	Polynomial
Crossover probability	0.9
Mutation probability	$\frac{1}{n}$

in *LCNQ* the set to approximate is larger, the output size was set to 6,000 members and the stopping criterion to 200,000 evaluations. In this benchmark, the variables to promote are x_3 and x_4 , i.e., two of the variables with little effect on the quality. Finally, $t_{do} = 0.04$ and $\epsilon = (0.01, 0.01)^\top$ were used and held constant for subsequent experiments unless otherwise stated.

Table II shows the median and standard deviation of the *MSTI* values obtained for both problems. For each problem, all the methods that are not statistically inferior to any other method are shown in bold face. The same applies to all the tables that show *MSTI* values in this paper. The median values attained by NSGA-II are significantly lower than the values obtained by other methods, so traditional MOEAs do not properly deal with approximating $N_{Q,\epsilon}$ even with incorporation of fast $N_{Q,\epsilon}$ and Algorithm 4. In the *DUNQ* problem, the most important benefits appear when the explicit management of diversity is activated. The second stage also contributes some additional benefits. In fact, the statistical tests show that VSD-MOEA-2S is superior to all the other approaches. Differently, in the case of *LCNQ*, most of the improvements are attributed to the activation of the second stage, whereas the explicit management of diversity is less important. Again, the statistical tests confirm the superiority of VSD-MOEA-2S against all the other approaches. As expected, the explicit management of diversity was the key to success for sets with unconnected regions, whereas the second phase is more important for large connected regions. The combination of both components provides an increased robustness since VSD-MOEA-2S obtained higher *MSTI* values and a low standard deviation in both problems. HREA offers advantages compared to NSGA-II and NSGA-II-2S in *DUNQ*, since promoting diversity is important. However, the explicit management of diversity used in VSD-MOEA, which is dependent on the stopping criterion, seems to be more suited for approximating $N_{Q,\epsilon}$. The results of HREA for *LCNQ* are not so promising. In this case, the main reason is that HREA lacks a component to promote diversity in specific variables. Instead, it promotes diversity in a global way (i.e., over the entire search space). The option of promoting diversity in specific variables is one of the contributions of TPEA- $N_{Q,\epsilon}$.

In order to visualize the results, MSTML plots are used. Figures 3 and 4 show these plots for *DUNQ* and *LCNQ*, respectively. In *DUNQ* it is noticeable that the methods based on VSD-MOEA are able to locate a higher number of distant regions. Methods based on NSGA-II place all the solutions in one or two of the regions, whereas HREA does not detect the

TABLE II
SUMMARY OF THE *MSTI* VALUES ATTAINED FOR *DUNQ* AND *LCNQ*

	<i>DUNQ</i>		<i>LCNQ</i>	
	Median	SD	Median	SD
NSGA-II	0.27	0.18	4.62	0.44
NSGA-II-2S	0.48	0.25	9.97	0.35
VSD-MOEA	0.94	0.07	6.79	0.35
VSD-MOEA-2S	1.09	0.06	10.59	0.24
HREA	0.53	0.06	4.47	0.20

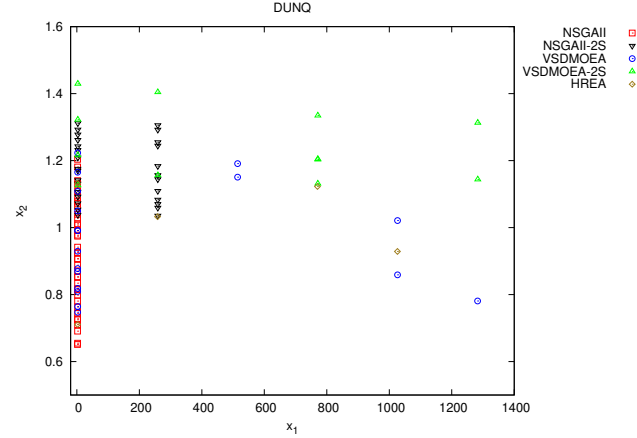


Fig. 3. MSTML plot for the *DUNQ* benchmark

most distant region. Moreover, VSD-MOEA-2S is not only able to identify most regions but also locates several points in each of the regions, which explains the larger *MSTI* value obtained by this method. In the case of the *LCNQ* none of the methods that consider only one phase operate properly. NSGA-II places most of the points in the bottom-left region, i.e., near to the region of the Pareto front. The evolutionary processes of VSD-MOEA and HREA promote diversity, resulting in an increased sampling of the solutions around the four corners. However, the central regions are undersampled. The activation of the second stage designed for TPEA- $N_{Q,\epsilon}$ provokes a much better spread of the solutions. In the case of NSGA-II-2S the second phase starts biased towards the bottom-left part. This is noticeable even at the end of the process, since the distance between solutions is shorter in such a region.

Since some large holes appear both in NSGA-II-2S and VSD-MOEA-2S and the second stage seems the key component to improve the spread, both methods were executed by setting $Stop_1$ to 10% of the total number of evaluations. This means that 180,000 evaluations are granted to the second stage. The MSTML plot for this case is shown in Figure 5. Here, there are no large regions uncovered, showing that the second stage is able to recover the diversity in large connected regions, obtaining a quite proper sampling of the desired region. The medians of the *MSTI* increased to 11.18 and 11.25, for NSGA-II-2S and VSD-MOEA-2S, respectively. Differences with respect to the previous settings were statistically significant, so while the second phase is useful to spread the solutions, it requires a somewhat large amount of resources, when an initially biased set is given.

For related formulations of multi-objective optimization,

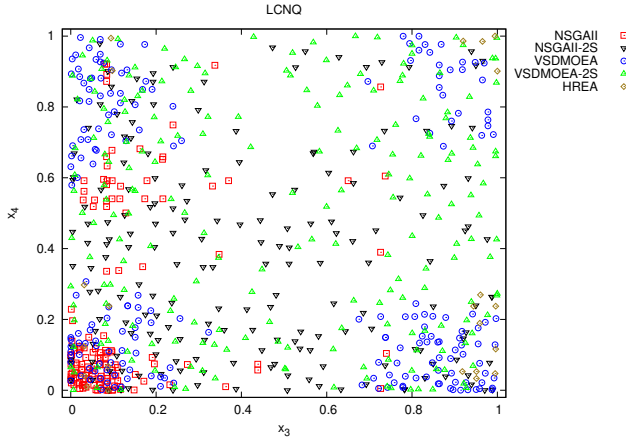


Fig. 4. MSTML plot for the LCNQ benchmark

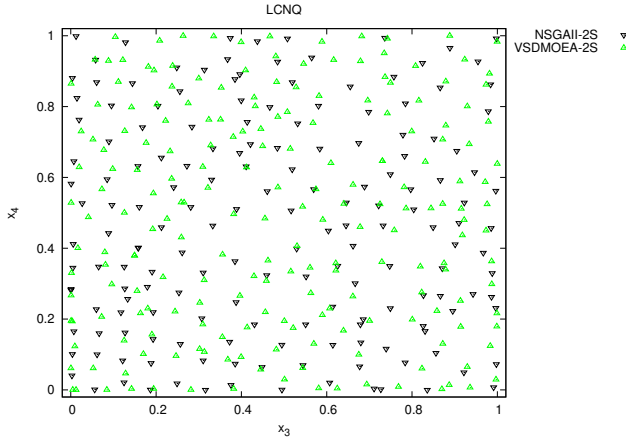


Fig. 5. MSTML plot for the LCNQ benchmark with a long second stage

such as multimodal multi-objective optimization, several benchmarks that consider unconnected or large regions with high-quality solutions have been defined. Thus, we selected five additional problems to confirm the trends detected with DUNQ and LCNQ. In particular, all the algorithms were executed with TMOP1-4 (see suppl. mat. for the definitions) and SYMPART9to9 (NPQe3) [12]. The value of ϵ_i was set to 0.1 for all the problems, except in NPQe3, for which the value 0.2 was used. Note that these settings were selected with the aim of creating interesting $N_{Q,\epsilon}$ sets, according to the penalties imposed to the different regions of the search space in each problem. Regarding the variables to promote, for problems with two variables, both variables are promoted, and for problems with three variables, the two last ones are promoted. The stopping criterion was set to 320,000 function evaluations and the desired output size was set to 300 for problems with two objectives and to 7,500 for problems with three objectives.

Table III shows the median MSTI values for all the approaches. VSD-MOEA-2S is the method that is highlighted in a larger number of problems. In fact, the only case where it was inferior to another method was in the problem TMOP1.

TABLE III
MEDIAN MSTI VALUES ATTAINED FOR BENCHMARKS USED IN MULTI-OBJECTIVE MULTIMODAL OPTIMIZATION.

	NSGA-II	NSGA-II-2S	VSD-MOEA	VSD-MOEA-2S	HREA
TMOP1	0.10	0.26	0.45	0.68	0.77
TMOP2	0.49	0.71	0.74	0.74	0.48
TMOP3	1.05	1.15	1.16	1.15	1.16
TMOP4	1.74	2.30	2.32	2.30	2.23
NPQe3	0.29	0.47	0.46	0.47	0.01

In this case, the way that HREA maintains the diversity resulted in a slightly superior performance. However, as we have mentioned previously, this method is not so robust in the sense that for some particular problems its performance is highly degraded. This happened for instance in NPQe3, where it obtained a value close to 0, while VSD-MOEA-2S attained 0.47. Regarding the behavior of the different realizations of $TPEA-N_{Q,\epsilon}$, for TMOP2, TMOP3, TMOP4 and NPQe3, both activating the second stage or applying VSD-MOEA are enough to reach high-quality solutions, i.e., activating both components simultaneously does not provide so important benefits. In this sense, these problems are easier than DUNQ and LCNQ. Differently, in the case of TMOP1, activating the second stage or VSD-MOEA alone provides some benefits but not as much as activating both simultaneously.

These additional results confirm the increased robustness of the way of managing the diversity included in VSD-MOEA for approximating $N_{Q,\epsilon}$, and the importance of including the second stage. While with VSD-MOEA or NSGA-II-2S several problems are solvable, there are also several other cases where they fail. The only method whose difference to the best one was not larger than 0.1 in all the problems was VSD-MOEA-2S. Thus, activating both the management of diversity and the second stage (VSD-MOEA-2S) is important to attain high-quality results in a robust way.

C. Convergence Speed of $TPEA-N_{Q,\epsilon}$

In the previous section, the importance of both stages of $TPEA-N_{Q,\epsilon}$ has been shown. The most promising setting is VSD-MOEA-2S, which uses VSD-MOEA in the first stage and activates the second stage. VSD-MOEA uses explicit management of diversity, which is a design principle that usually results in competitive but computationally demanding algorithms. Moreover, the second stage is also demanding since it uses large population sizes, so the number of evaluations highly increases with the number of generations. Thus, we expected this variant of $TPEA-N_{Q,\epsilon}$ to excel especially in the long term.

While the previous section has shown the effectiveness of VSD-MOEA-2S, this section focuses on the efficiency. In particular, we are interested in analyzing the performance of VSD-MOEA-2S under different computational capabilities. All the algorithms were executed 30 times by setting the stopping criterion to 10,000, 25,000, 100,000, 200,000 and 320,000 function evaluations with the 7 problems studied in the previous section.

In order to summarize the results, the maximum MSTI obtained by any of the methods for each problem was calculated. This value was used to normalize the MSTI values,

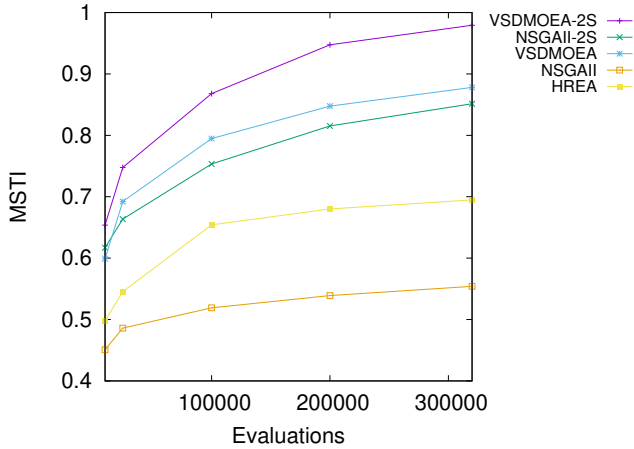


Fig. 6. Trend of the MSTI values considering normalized data for 7 problems

TABLE IV
SUMMARY OF THE RESULTS OF THE STATISTICAL TESTS FOR DIFFERENT STOPPING CRITERIA

	10K		25K		100K		200K		320K	
	↑	↓	↑	↓	↑	↓	↑	↓	↑	↓
VSD-MOEA-2S	17	1	19	1	18	1	18	2	19	1
VSD-MOEA	9	11	12	6	13	7	15	7	15	7
NSGA-II-2S	14	5	10	7	11	6	11	6	13	7
HREA	7	13	6	12	7	13	7	13	7	15
NSGA-II	3	20	1	22	2	24	1	24	1	25

since the ranges of MSTI vary depending on the problem. Figure 6 shows the mean of the normalized MSTI values over the seven problems. In the short term VSD-MOEA-2S is the method that attains the largest MSTI values. However, its advantage against the other methods is not so outstanding. As more evaluations are granted to the methods, the superiority of VSD-MOEA-2S is clearer. Thus, as expected, situations where long runs are admissible are the cases where VSD-MOEA-2S excel. However, even with shorter executions it remains competitive. Regarding the proper employment of increased computational resources, methods that include the explicit management of diversity (VSD-MOEA) or activate a second stage employ them better, resulting in a larger slope. It is also noticeable that even with 320,000 functions evaluations, the MSTI value keeps increasing for VSD-MOEA-2S. Thus, in order to attain fine-grained approximations of $N_{Q,\epsilon}$, a large amount of computational resources are required. However, using much less evaluations might be suitable in some cases, since relatively large MSTI values can also be attained with VSD-MOEA-2S in shorter executions.

In order to get a deeper understanding, pairwise statistical tests were performed among the methods for each stopping criterion and problem tested. Table IV shows, for each stopping criterion, the amount of comparisons where each method was statistically superior (↑) and inferior (↓) in terms of MSTI values. For every stopping criterion, the superiority of VSD-MOEA-2S is clear. From a total of 140 tests, it was superior in 91 cases and inferior in just 6 cases. The inferiority of NSGA-II is clearer in the long term, since it converges and it is not able to explore additional regions. It is also clear

that incorporating the management of diversity alone (VSD-MOEA) is more beneficial in long-term settings. Taking all these results into account, it can be concluded that both phases are important for proper performance, and in particular, the second stage is more important for short-term settings than for long-term settings. VSD-MOEA-2S is the most competitive approach independently of the computational budget.

D. Performance in MOPs with More Complex Features

Previous sections have analyzed problems that have some features that make them specially suited for solvers that search for nearly optimal solutions. However, they lack some complexities usually considered in the analysis of MOEAs, such as bias or deception, among others. This section analyzes the behavior of TPEA- $N_{Q,\epsilon}$ with a complex set of problems that presents a variety of features: the WFG benchmark. Standard parameters for configuring the WFG benchmark were used. Particularly, 20 distance-related and four position-related variables are considered in all the problems. Both two-objective and three-objective cases were considered. Similarly to the previous section, NSGA-II, NSGA-II-2S, VSD-MOEA and VSD-MOEA-2S were tested. Preliminary executions were also performed with HREA. However, since 24 variables were used and HREA lacks the option of promoting diversity in specific subsets of variables, MSTI values were close to 0, so it is not included in this study. The parameterization of the optimizers was also maintained (see Table I) and each component of ϵ was set to 0.01. Since the functions are much more complex, the stopping criterion was set to 750,000 function evaluations. The output size was set to 300 and 7,500 for the two- and three-objective cases, respectively. Two variables, 23 and 24, were selected for diversity promotion. Note that these two variables are distance variables. We chose to promote diversity in distance variables because, unlike position variables, their diversity is not implicitly promoted by the objective space spread. Thus, maintaining diversity in these variables is more challenging than doing so for position variables.

Tables V and VI summarize the MSTI values (median and standard deviation) obtained for every problem with two and three objectives, respectively. Differently to previous sections, the performance of VSD-MOEA-2S and NSGA-II-2S is quite similar in most cases. The reason is that in most cases, the identified approximation of $N_{Q,\epsilon}$ set is connected and appears in a relatively small region. Thus, applying NSGA-II-2S is enough to attain a proper approximation in such cases. Thus, the advantages of using VSD-MOEA in the first phase is not so evident as in previous experiments. However, there are two clear exceptions: WFG5 and WFG8. WFG5 has a disconnected $N_{Q,\epsilon}$ set because it is a deceptive problem in which, for each position-variable, three different regions are promising. VSD-MOEA is able to detect the different regions resulting in an increased MSTI value and the second phase aids to better spread the solutions in each of these regions. WFG8 present complex dependencies among the variables but VSD-MOEA can solve it [56]. However, a lot of evaluations are required, so integrating the second stage is detrimental. Further interesting cases to discuss are WFG1, WFG6 and

TABLE V
SUMMARY OF THE MSTI VALUES ATTAINED FOR WFG WITH TWO OBJECTIVES.

	NSGA-II		NSGA-II-2S		VSD-MOEA		VSD-MOEA-2S	
	Med.	SD	Med.	SD	Med.	SD	Med.	SD
WFG1	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
WFG2	0.38	0.02	0.64	0.01	0.36	0.08	0.64	0.15
WFG3	0.62	0.02	1.10	0.03	0.58	0.02	1.10	0.02
WFG4	1.09	0.04	1.87	0.04	1.00	0.06	1.87	0.04
WFG5	0.61	0.46	1.58	0.84	2.70	0.30	2.95	0.08
WFG6	0.14	0.17	0.21	0.24	0.00	0.00	0.00	0.00
WFG7	0.87	0.01	1.15	0.02	0.85	0.03	1.14	0.02
WFG8	0.05	0.04	0.03	0.03	0.24	0.03	0.14	0.07
WFG9	0.00	0.13	0.00	0.22	0.00	0.12	0.00	0.10

WFG9. In WFG1 and WFG9, the median of the MSTI is 0 in every method. WFG1 is a biased problem, meaning that very small changes in the variables provoke drastic changes in the objective space. Thus, it presents a tiny $N_{Q,\epsilon}$ set. In the case of WFG6 and WFG9, variables are non-separable. This results in a really slow convergence with crossover operators that do not incorporate learning strategies. In fact, it has been shown that SBX keeps steadily improving the convergence to the Pareto front even after 250,000 generations [58]. Thus, none of the methods is robust enough to get proper approximations. In the case of WFG9, the results are not robust against different seeds, resulting in a median MSTI value equal to 0 in every method. In the case of WFG6, the seed does not heavily affects the results. Thus, in spite that none of the methods converges to the Pareto front, the approximations of NSGA-II are always better than those of VSD-MOEA because of the increased elitism. Interestingly, WFG6 has a disconnected $N_{Q,\epsilon}$ set and VSD-MOEA-2S identify solutions in each of the regions. However, this is done at the cost of a further reduction in the velocity of convergence towards the Pareto front. Thus, WFG6 represents a case where combining our strategy with advanced crossover operators [59] might be helpful. This combination, however, is beyond the scope of this research.

The experimentation with WFG remarks the important benefits of activating the second stage. However, it also shows some limitations. Due to the explicit increase of diversity promoted by VSD-MOEA, the convergence speed towards the Pareto front is reduced. If the stopping criterion is such that the method is not able to converge, this might result in a reduced performance against more elitist methods. Thus, our recommendation is to use VSD-MOEA when we suspect that $N_{Q,\epsilon}$ is disconnected and the DM is able to accept the increase in the demands of computational resources. Particularly, the amount of resources devoted to the first stage should be large enough to ensure that the Pareto front is properly approximated with a large probability. Otherwise, more elitist methods might be preferred for the first phase.

E. Effect of ϵ and t_{do} on the decision-making process

Depending on the outputs desired by the DM, different ϵ and t_{do} are required. This section discusses the effect of both parameters in a practical case. The problem considered is a slight modification of SSW [60], referred to as SSW2 in the sequel, that arises in product design: variable x_i , $i = 1, \dots, 5$, represents the amount of material m_i used for the product.

TABLE VI
SUMMARY OF THE MSTI VALUES ATTAINED FOR WFG WITH THREE OBJECTIVES.

	NSGA-II		NSGA-II-2S		VSD-MOEA		VSD-MOEA-2S	
	Med.	SD	Med.	SD	Med.	SD	Med.	SD
WFG1	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
WFG2	0.31	0.07	0.68	0.14	0.26	0.03	0.71	0.06
WFG3	1.01	0.11	2.12	0.01	1.81	0.11	2.11	0.01
WFG4	0.69	0.03	1.07	0.06	0.62	0.03	1.34	0.03
WFG5	0.44	0.29	1.55	0.51	2.11	0.51	2.71	0.04
WFG6	0.26	0.08	0.87	0.35	0.02	0.03	0.01	0.01
WFG7	0.60	0.01	0.94	0.01	0.50	0.02	1.04	0.01
WFG8	0.07	0.01	0.07	0.01	0.15	0.01	0.25	0.04
WFG9	0.00	0.15	0.00	0.21	0.00	0.00	0.00	0.00

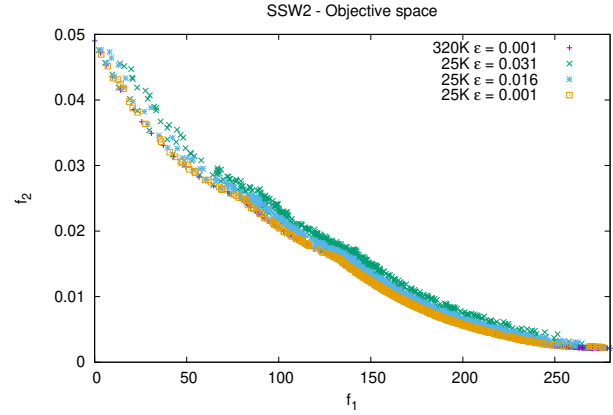


Fig. 7. Objective space visualization of runs that obtained the median MSTI for different ϵ and stopping criteria

The first objective is the cost, and the second represents the failure rate. Refer to the suppl. mat. for the formal definition and for a decision-making example using this problem.

In this section, we assume that diversity is to be promoted in the first two variables ($VP = \{x_1, x_2\}$). This might be useful for properly dealing with potential shortage of these materials as is shown in the suppl. mat. First, we analyze the effect of the ϵ parameter. The same value is used in each component of the vector, so $\epsilon = v$ is a shortcut for $\epsilon = (v, v)^T$ in this section. To this end, SSW2 was solved with VSD-MOEA-2S using the same parameterization as in previous experiments and $\epsilon \in \{0.001, 0.016, 0.031\}$. We used 25,000 evaluations, but also included the results with $\epsilon = 0.001$ using 320,000 evaluations for comparison purposes. The output size (N_2) was set to 600 and $t_{do} = 0.04$. Figure 7 shows the results of the runs that achieved the median MSTI for each configuration. The effect of ϵ is as expected. With $\epsilon = 0.001$ — a very small value —, most solutions are non-dominated. Moreover, the results are quite similar regardless of the stopping criterion, which indicates that 25,000 evaluations are enough to reach high-quality solutions. As ϵ increases, a larger level of deterioration is accepted. These kinds of visualizations can assist the DM in deciding on the level of acceptable deterioration.

Regarding the variable space, we present both the final set obtained by the runs that achieved the median MSTI and the corresponding MSTML plots. Figure 8 shows the final sets in the variable space, showing that extended executions are beneficial for increasing diversity. This is exemplified for

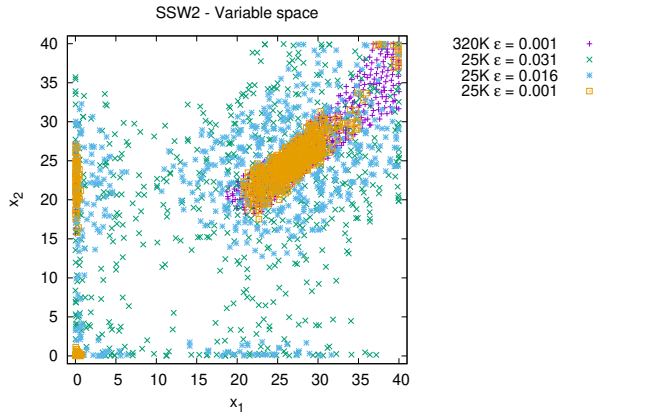


Fig. 8. Variable space visualization of runs that obtained the median MSTI for different ϵ and stopping criteria

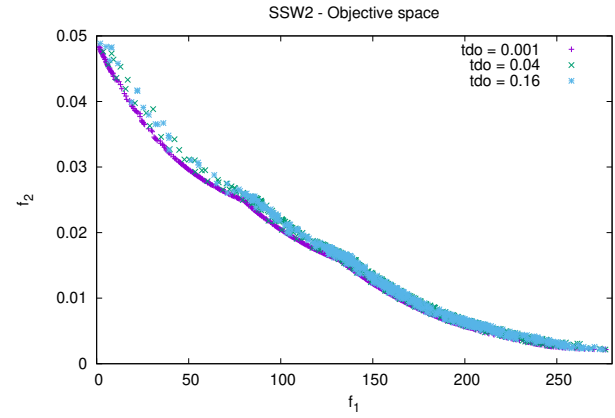


Fig. 10. Objective space visualization of runs that obtained the median MSTI for different tdo with $\epsilon = 0.016$ and 25,000 evaluations

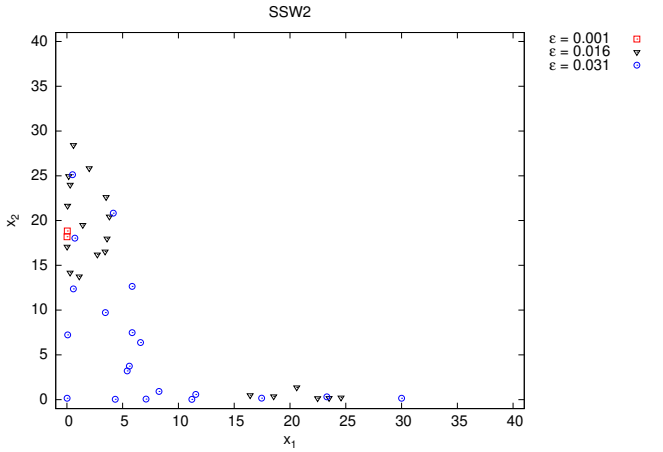


Fig. 9. Effect of ϵ in the MSTML plots for the SSW2 problem. Executions with 25,000 evaluations are considered.

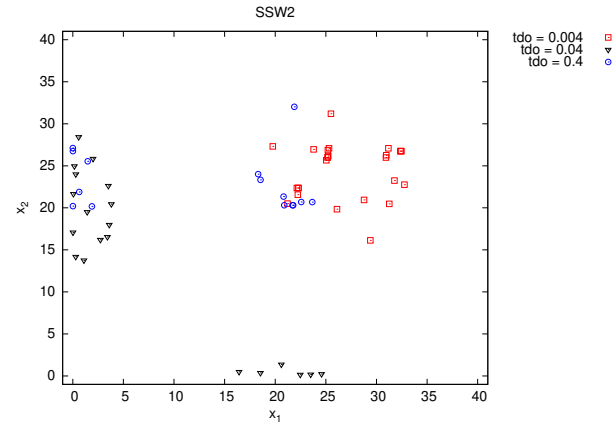


Fig. 11. MSTML plots for different tdo with $\epsilon = 0.016$ and 25,000 evaluations

$\epsilon = 0.001$; similar results were observed for other values of ϵ , but are omitted to avoid cluttering the figure. The execution with 25,000 evaluations and $\epsilon = 0.001$ identifies the same three regions as the longer runs but with a lower spread. Thus, preliminary fast runs can be conducted to effectively support the DM's process. Concerning the effect of ϵ , increasing its value to 0.016 enlarges the regions and identifies new areas of interest. Finally, when $\epsilon = 0.031$ is used, the regions expand to the point that some of them are merged, resulting in a significant increase in diversity.

Regarding the MSTML plots shown in Figure 9, the advantages of increasing ϵ are evident. With $\epsilon = 0.001$, only small trees are identified. By increasing ϵ to 0.016, two regions are covered, and with the largest value of ϵ areas that connect these two regions can also be utilized, thus presenting a wider range of alternative solutions to the DM.

Regarding the tdo parameter, in addition to the value 0.04 used in all previous experiments, we tested 0.001 and 0.16 to analyze the effects of small and large values, respectively. Figure 10 shows the output generated by each configuration in the objective space with $\epsilon = 0.016$. When tdo is set

to a very small value, the bounded selector (Algorithm 4) selects individuals by only considering the objectives, i.e., by maximizing diversity among non-dominated solutions. Consequently, it approximates the Pareto front. In the case of using $tdo = 0.16$, the output is quite similar to the one obtained with $tdo = 0.04$. Under the configuration $tdo = 0.16$, only a few solutions are selected by maximizing objective-space distances, while the remainder are selected by maximizing diversity in the variable space. However, maximizing variable-space diversity within $N_{Q,\epsilon}$ also leads to proper diversity in the objective space provided there is a significant correlation between the diversity of the variable space and that of the objective space. Thus, the deterioration is lower than with very small tdo values.

In terms of MSTI, configurations with $tdo \in \{0.001, 0.04, 0.16\}$ obtained values of 0.13, 1.06, and 0.91. This confirms that while deterioration occurred when using large tdo values, it was not significant. Figure 11 shows the MSTML plots for these cases. In the median case, $tdo \in \{0.04, 0.16\}$ cover two regions — with a wider extension with $tdo = 0.04$ —, whereas using $tdo = 0.001$ provides the DM with alternatives in only one region.

VI. CONCLUSIONS AND FUTURE WORK

In this work, we have made a first effort to develop evolutionary algorithms for the computation of the set $N_{Q,\epsilon}$ of a given multi-objective optimization problem (MOP). This set contains all optimal solutions (i.e., the entire Pareto set) as well as non- but nearly optimal solutions. The entire set is hence relevant for the decision maker (DM) of a given problem since it contains all optimal solutions as well as further (non-optimal) ones that may still serve as alternative or back-up solutions for the realization of the project. We have proposed the two-phase framework TPEA- $N_{Q,\epsilon}$. In the first phase, the focus of the algorithm is to obtain an approximation of the Pareto set. However, all points in $N_{Q,\epsilon}$ detected in this phase will be used as initial candidates for the second phase which focuses on non- but nearly optimal solutions. The second phase incorporates a clustering-based bounded selector in the replacement strategy and operates with larger populations. In order to manage efficiently the large sets considered in TPEA- $N_{Q,\epsilon}$, a fast $N_{D,\epsilon}$ algorithm for detecting all nearly optimal solutions of a discrete set D of candidate solutions is proposed. Similar to the best non-dominated detection algorithms, its complexity is $O(|D| \log |D|)$ for two and three-objective problems. However, it requires more complex data-structures and methodologies such as segment trees and the two-pointer technique. Note that the straightforward detection of $N_{D,\epsilon}$ has a complexity of $O(N^2)$, which is too slow for the experimentation carried out in this research. In addition to the proposals, a methodology to compare $N_{Q,\epsilon}$ approximations is defined, which includes both indicators and visualization techniques. The experimental validation clearly shows the important benefits of using explicit techniques to manage the diversity of the population in the first phase, as well as the incorporation of the clustering-based replacement. The comparisons included complex well-known MOPs, problems designed for multimodal multi-objective optimization, two newly defined problems and a design problem concerning material selection. The experimental validation shows that TPEA- $N_{Q,\epsilon}$ with VSD-MOEA and two phases has an outstanding performance in the long-term. Additionally, it can also be used in the short-term to attain proper approximations.

One of the main weaknesses detected is that, specially for the most complex problems, the amount of evaluations required to obtain fine-grained approximations of $N_{Q,\epsilon}$ is large. During the second phase, large population are managed to approximate a large set. Thus, the parent selection operators might be modified by including assortative mating restrictions. Additionally, the operators might be modified with the aim of oversampling the regions where larger holes are identified, which might probably require adaptive techniques, because the shape of $N_{Q,\epsilon}$ is usually unknown. In order to reduce the memory requirement of TPEA- $N_{Q,\epsilon}$, dynamic archivers for maintaining $N_{D,\epsilon}$ should be defined. Another interesting avenue consists in enhancing the anytime behavior of the proposed approach to facilitate interaction with DMs. In such settings, algorithms that move away from two distinct, sequential phases are likely more effective. Furthermore, we note that effective interaction with the DM requires both

further formalization and the development of front-end tools, representing a promising direction for future research.

CONFLICT OF INTEREST

The authors declare that they have no conflict of interest.

REFERENCES

- [1] K. Deb, *Multi-Objective Optimization using Evolutionary Algorithms*. Chichester, UK: John Wiley & Sons, 2001, ISBN 0-471-87339-X.
- [2] J. Liang, X. Ban, K. Yu, B. Qu, K. Qiao, C. Yue, K. Chen, and K. C. Tan, "A survey on evolutionary constrained multiobjective optimization," *IEEE Transactions on Evolutionary Computation*, vol. 27, no. 2, pp. 201–221, 2023.
- [3] C. A. Coello Coello, E. Goodman, K. Miettinen, D. Saxena, O. Schütze, and L. Thiele, "Interview: Kalyanmoy Deb talks about formation, development and challenges of the EMO community, important positions in his career, and issues faced getting his works published," *Mathematical and Computational Applications*, vol. 28, no. 2, 2023.
- [4] H. Ma, Y. Zhang, S. Sun, T. Liu, and Y. Shan, "A comprehensive survey on NSGA-II for multi-objective optimization and applications," *Artificial Intelligence Review*, vol. 56, p. 15217–15270, 2023.
- [5] D. Molina-Pérez and A. G. Rojas-López, "Resolving contrast and detail trade-offs in image processing with multi-objective optimization," *Mathematical and Computational Applications*, vol. 29, no. 6, 2024.
- [6] G. Goni, S. Nesmachnow, D. Rossit, P. Moreno-Bernal, and A. Tchermnykh, "Bio-inspired multiobjective optimization for designing content distribution networks," *Mathematical and Computational Applications*, vol. 30, no. 2, 2025.
- [7] G. Rudolph and M. Preuss, "A multiobjective approach for finding equivalent inverse images of Pareto-optimal objective vectors," in *2009 IEEE Symposium on Computational Intelligence in Multi-Criteria Decision-Making (MCDM)*. IEEE, 2009, pp. 74–79.
- [8] C. Yue, B. Qu, and J. Liang, "A multiobjective particle swarm optimizer using ring topology for solving multimodal multiobjective problems," *IEEE Transactions on Evolutionary Computation*, vol. 22, no. 5, pp. 805–817, 2017.
- [9] R. Tanabe and H. Ishibuchi, "A review of evolutionary multimodal multiobjective optimization," *IEEE Transactions on Evolutionary Computation*, vol. 24, no. 1, pp. 193–200, 2019.
- [10] A. Ahrari, S. Elsayed, R. Sarker, D. Essam, and C. A. Coello Coello, "Adaptive multilevel prediction method for dynamic multimodal optimization," *IEEE Transactions on Evolutionary Computation*, vol. 25, no. 3, pp. 463–477, 2021.
- [11] W. Li, X. Yao, T. Zhang, R. Wang, and L. Wang, "Hierarchy ranking method for multimodal multiobjective optimization with local Pareto fronts," *IEEE Transactions on Evolutionary Computation*, vol. 27, no. 1, pp. 98–110, 2022.
- [12] A. E. Rodríguez-Fernández, L. Schäpermeier, C. Hernández, P. Kerschke, H. Trautmann, and O. Schütze, "Finding ϵ -locally optimal solutions for multi-objective multimodal optimization," *IEEE Transactions on Evolutionary Computation*, pp. 1–1, 2024.
- [13] O. Schütze, C. A. Coello, and E.-G. Talbi, "Approximating the ϵ -efficient set of an mop with stochastic search algorithms," in *MICAI 2007*, A. Gelbukh and A. F. K. Morales, Eds., Berlin, Heidelberg, 2007, pp. 128–138.
- [14] O. Schütze, C. Hernández, E.-G. Talbi, J. Q. Sun, Y. Naranjani, and F. R. Xiong, "Archivers for the representation of the set of approximate solutions for MOPs," *Journal of Heuristics*, vol. 5, no. 1, pp. 71 – 105, 2019.
- [15] O. Schütze, A. E. Rodríguez-Fernández, C. Segura, and C. Hernández, "Finding the set of nearly optimal solutions of a multi-objective optimization problem," *IEEE Transactions on Evolutionary Computation*, pp. 1–1, 2024.
- [16] P. Lordin, " ϵ -solutions in vector minimization problems," *Journal of Optimization, Theory and Application*, vol. 42, pp. 265–276, 1984.
- [17] D. J. White, "Epsilon efficiency," *Journal of Optimization Theory and Applications*, vol. 49, no. 2, pp. 319–337, 1986.
- [18] M. Laumanns, L. Thiele, K. Deb, and E. Zitzler, "Combining convergence and diversity in evolutionary multiobjective optimization," *Evolutionary computation*, vol. 10, pp. 263 – 282, 2002.
- [19] K. Deb, A. Pratap, S. Agarwal, and T. Meyarivan, "A fast and elitist multiobjective genetic algorithm: NSGA-II," *Evolutionary Computation, IEEE Transactions on*, vol. 6, no. 2, pp. 182–197, 2002.

- [20] E. Zitzler, M. Laumanns, and L. Thiele, "SPEA2: Improving the Strength Pareto Evolutionary Algorithm for Multiobjective Optimization," in *Evolutionary Methods for Design, Optimisation and Control with Application to Industrial Problems (EUROGEN 2001)*, K. Giannakoglou et al., Eds. International Center for Numerical Methods in Engineering (CIMNE), 2002, pp. 95–100.
- [21] Q. Zhang and H. Li, "MOEA/D: A multi-objective evolutionary algorithm based on decomposition," *IEEE Transactions on Evolutionary Computation*, vol. 11, no. 6, pp. 712–731, 2007.
- [22] K. Deb and H. Jain, "An evolutionary many-objective optimization algorithm using reference-point-based nondominated sorting approach, part I: Solving problems with box constraints," *Transactions on Evolutionary Computation*, vol. 18, no. 4, pp. 577–601, 2014.
- [23] N. Beume, B. Naujoks, and M. Emmerich, "SMS-EMOA: Multiobjective selection based on dominated hypervolume," *European Journal of Operational Research*, vol. 181, no. 3, pp. 1653–1669, 2007.
- [24] S. Zapotecas-Martínez, A. López-Jaimes, and A. García-Nájera, "LIBEA: A Lebesgue indicator-based evolutionary algorithm for multi-objective optimization," *Swarm and Evolutionary Computation*, vol. 44, pp. 404 – 419, 2019.
- [25] E. Zitzler and L. Thiele, "Multiobjective evolutionary algorithms: A comparative case study and the strength Pareto approach," *IEEE Transactions on Evolutionary Computation*, vol. 3, no. 4, pp. 257–271, 1999.
- [26] C. A. R. Villalobos and C. A. C. Coello, "A new multi-objective evolutionary algorithm based on a performance assessment indicator," in *Proceedings of the 14th Annual Conference on Genetic and Evolutionary Computation*, 2012, p. 505–512.
- [27] O. Schütze, X. Esquivel, A. Lara, and C. A. Coello Coello, "Using the averaged Hausdorff distance as a performance measure in evolutionary multi-objective optimization," *IEEE Transactions on Evolutionary Computation*, vol. 16, no. 4, pp. 504–522, 2012.
- [28] O. Schütze and C. Hernández, *Archiving Strategies for Evolutionary Multi-objective Optimization Algorithms*. Springer, 2021.
- [29] J. E. Fieldsend, "Using an adaptive collection of local evolutionary algorithms for multi-modal problems," *Soft Computing*, vol. 19, 06 2014.
- [30] J. E. Fieldsend and K. Alyahya, "Visualising the landscape of multi-objective problems using local optima networks," in *Proceedings of the Genetic and Evolutionary Computation Conference Companion*, 2019, pp. 1421–1429.
- [31] L. Schäpermeier, C. Grimme, and P. Kerschke, "One PLOT to Show Them All: Visualization of Efficient Sets in Multi-objective Landscapes," in *International Conference on Parallel Problem Solving from Nature*. Springer, 2020, pp. 154–167.
- [32] C. Grimme, P. Kerschke, P. Aspar, H. Trautmann, M. Preuss, A. H. Deutz, H. Wang, and M. Emmerich, "Peeking beyond peaks: Challenges and research potentials of continuous multimodal multi-objective optimization," *Computers & Operations Research*, vol. 136, p. 105489, 2021.
- [33] E. M. Zechman and S. R. Ranjithan, "An evolutionary algorithm to generate alternatives (EAGA) for engineering optimization problems," *Engineering Optimization*, vol. 36, no. 5, pp. 539–553, 2004. [Online]. Available: <https://doi.org/10.1080/03052150410001704863>
- [34] T. Ulrich, J. Bader, and E. Zitzler, "Integrating decision space diversity into hypervolume-based multiobjective search," in *Proceedings of the 12th Annual Conference on Genetic and Evolutionary Computation*, ser. GECCO '10. New York, NY, USA: Association for Computing Machinery, 2010, p. 455–462. [Online]. Available: <https://doi.org/10.1145/1830483.1830569>
- [35] K. Pereverdieva, A. Deutz, T. Ezendam, T. Bäck, H. Hofmeyer, and M. Emmerich, "Comparative analysis of indicators for multi-objective diversity optimization," in *Evolutionary Multi-Criterion Optimization*, H. Singh, T. Ray, J. Knowles, X. Li, J. Branke, B. Wang, and A. Oyama, Eds. Singapore: Springer Nature Singapore, 2025, pp. 58–71.
- [36] H. T. Kung, F. Luccio, and F. P. Preparata, "On finding the maxima of a set of vectors," *Journal of the ACM*, vol. 22, no. 4, pp. 469–476, 1975.
- [37] F. P. Preparata and M. I. Shamos, *Computational geometry: an introduction*. Berlin, Heidelberg: Springer-Verlag, 1985.
- [38] M. T. Jensen, "Reducing the run-time complexity of multiobjective EAs: The NSGA-II and other algorithms," *IEEE Transactions on Evolutionary Computation*, vol. 7, no. 5, pp. 503–515, 2003.
- [39] F.-A. Fortin, S. Grenier, and M. Parizeau, "Generalizing the improved run-time complexity algorithm for non-dominated sorting," in *Proceedings of the 15th annual conference on Genetic and evolutionary computation*, 2013, pp. 615–622.
- [40] S. Mishra, S. Saha, S. Mondal, and C. A. C. Coello, "A divide-and-conquer based efficient non-dominated sorting approach," *Swarm and evolutionary computation*, vol. 44, pp. 748–773, 2019.
- [41] J. Moreno, D. Rodriguez, A. J. Nebro, and J. A. Lozano, "Merge nondominated sorting algorithm for many-objective optimization," *IEEE Transactions on Cybernetics*, vol. 51, no. 12, pp. 6154–6164, 2020.
- [42] Q. Long, X. Wu, and C. Wu, "Non-dominated sorting methods for multi-objective optimization: Review and numerical comparison," *Journal of Industrial & Management Optimization*, vol. 17, no. 2, 2021.
- [43] S. Mishra, V. Prakash, and C. A. Coello Coello, "Analysis of merge non-dominated sorting algorithm," in *International Conference on Evolutionary Multi-Criterion Optimization*. Springer, 2025, pp. 46–57.
- [44] R. Denysiuk, L. Costa, and I. Espírito Santo, "Clustering-based selection for evolutionary many-objective optimization," in *Parallel Problem Solving from Nature – PPSN XIII*, T. T. Bartz-Beielstein, J. Branke, B. Filipič, and J. Smith, Eds., 2014, pp. 538–547.
- [45] E. Zitzler and L. Thiele, "Multiobjective evolutionary algorithms: A comparative case study and the strength Pareto approach," *IEEE Transactions on Evolutionary Computation*, vol. 3, no. 4, pp. 257–271, 1999.
- [46] Zitzler, Eckart, Laumanns, Marco, and Thiele, Lothar, "SPEA2: Improving the strength pareto evolutionary algorithm," Tech. Rep., 2001. [Online]. Available: <http://hdl.handle.net/20.500.11850/145755>
- [47] H. K. Singh, K. S. Bhattacharjee, and T. Ray, "Distance-based subset selection for benchmarking in evolutionary multi/many-objective optimization," *IEEE Transactions on Evolutionary Computation*, vol. 23, no. 5, pp. 904–912, 2019.
- [48] K. Shang, H. Ishibuchi, and Y. Nan, "Distance-based subset selection revisited," in *Proceedings of the Genetic and Evolutionary Computation Conference*, ser. GECCO '21, 2021, p. 439–447.
- [49] K. Shang, T. Shu, H. Ishibuchi, Y. Nan, and L. M. Pang, "Benchmarking large-scale subset selection in evolutionary multi-objective optimization," *Information Sciences*, vol. 622, pp. 755–770, 2023.
- [50] W. Chen, H. Ishibuchi, and K. Shang, "Clustering-based subset selection in evolutionary multiobjective optimization," in *2021 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, 2021, pp. 468–475.
- [51] J. MacQueen, "Some methods for classification and analysis of multivariate observations," in *Proceedings of the 5th Berkeley Symposium on Mathematical Statistics and Probability - Vol. 1*, L. M. Le Cam and J. Neyman, Eds. University of California Press, Berkeley, CA, USA, 1967, pp. 281–297.
- [52] L. Uribe, J. M. Bogoya, A. Vargas, A. Lara, G. Rudolph, and O. Schütze, "A set based Newton method for the averaged Hausdorff distance for multi-objective reference set problems," *Mathematics*, vol. 8, no. 10, 2020.
- [53] M. Geilen and T. Basten, "A calculator for Pareto points," in *2007 Design, Automation & Test in Europe Conference & Exhibition*, 2007, pp. 1–6.
- [54] P. Brass, *Advanced Data Structures*. Cambridge University Press, 2008.
- [55] A. Laaksonen, *Guide to Competitive Programming: Learning and Improving Algorithms Through Contests*. Springer, 2020.
- [56] J. Chacón Castillo, C. Segura, and C. A. C. Coello, "VSD-MOEA: A Dominance-Based Multiobjective Evolutionary Algorithm with Explicit Variable Space Diversity Management," *Evolutionary Computation*, vol. 30, no. 2, pp. 195–219, 2022.
- [57] I. G. del Amo and D. A. Pelta, "SRCS: a technique for comparing multiple algorithms under several factors in dynamic optimization problems," in *Metaheuristics for Dynamic Optimization*. Berlin, Heidelberg: Springer, 2013, pp. 61–77.
- [58] S. Huband, P. Hingston, L. Barone, and L. While, "A review of multiobjective test problems and a scalable test problem toolkit," *IEEE Transactions on Evolutionary Computation*, vol. 10, no. 5, pp. 477–506, 2006.
- [59] J. Heise and S. Mostaghim, "Online learning hyper-heuristics in multi-objective evolutionary algorithms," in *Evolutionary Multi-Criterion Optimization*, M. E. et al., Ed., 2023, pp. 162–175.
- [60] S. Schaeffler, R. Schultz, and K. Weinzierl, "Stochastic method for the solution of unconstrained vector optimization problems," *Journal of Optimization Theory and Applications*, vol. 114, no. 1, pp. 209–222, 2002.